AQI Estimation from Sky Images

Mini-Project Report

Comparing Custom CNN | CNN + Random Forest | CNN + SVM (SVR)

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1. Introduction

Air Quality Index (AQI) is a standardized metric that quantifies the degree of air pollution. Accurate real-time AQI estimation is critical for public health advisories, urban planning, and environmental monitoring. Traditional ground-based sensors are expensive and geographically sparse, creating a need for scalable, low-cost estimation alternatives.

This project investigates whether sky photographs — captured by ordinary cameras or smartphones — can serve as a proxy for AQI estimation. Haze, scattering, and colour shifts in sky images are visually correlated with particulate matter (PM2.5) concentrations, which directly map to the AQI scale.

Three machine-learning pipelines are designed and evaluated:

- Model 1 — Custom CNN: An end-to-end convolutional neural network trained directly to regress AQI from sky images.
- Model 2 — CNN + Random Forest: The trained CNN is used as a frozen feature extractor; its 512-dimensional Global Average Pooling (GAP) embeddings feed a Random Forest regressor.
- Model 3 — CNN + SVM (SVR): The same 512-d CNN embeddings feed a Support Vector Regressor (RBF kernel), with Standard Scaler normalisation.

The report documents each model's design, training procedure, and quantitative performance using MAE, RMSE, R^2 score, and AQI-category accuracy.

2. Dataset & Preprocessing

You can find my dataset in
https://drive.google.com/drive/folders/1fNp0J51e-eYf5-jtbzhxBsGPF0PJxtNt?usp=drive_link

2.1 Data Description

The dataset consists of labelled sky images paired with PM2.5 readings stored in a CSV file (labels.csv). Only images with valid filenames and AQI values in the range [0, 500] are retained after cleaning.

- Images: RGB sky photographs resized to 224 × 224 pixels.
- Labels: PM2.5-based AQI values (continuous, range 0–500).
- AQI categories: Good (0–50), Moderate (51–100), Unhealthy for Sensitive Groups (101–150), Unhealthy (151–200), Very Unhealthy (201–300), Hazardous (301–500).

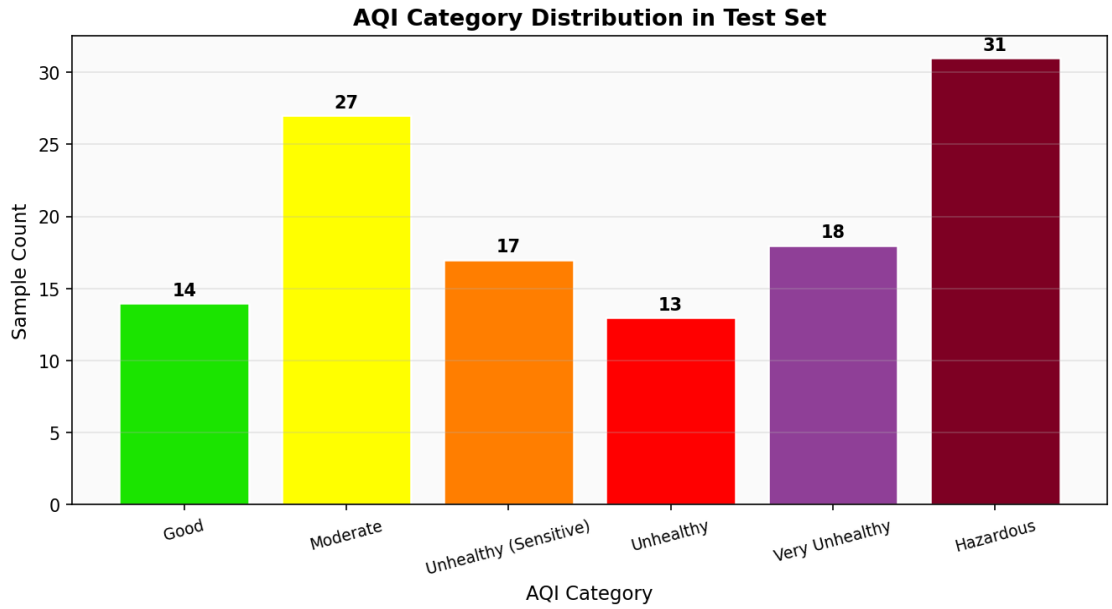

Figure 1 — AQI category distribution of the test set. Most samples fall in low-to-moderate pollution ranges.

2.2 Data Splits

A stratified 70 / 15 / 15 split is applied with **random seed = 42** for reproducibility:

- Training set: 70% of labelled images — used for CNN weight optimisation.
- Validation set: 15% — used for learning-rate scheduling and early stopping.
- Test set: 15% — held out for all final metric evaluation.

For the sklearn models (RF, SVM), training and validation sets are merged (train + val) to maximise the data available for hyperparameter tuning via 3-fold cross-validation.

2.3 Image Preprocessing & Augmentation

Sky colour is the primary AQI signal, so colour augmentation is kept minimal to avoid distorting the pollution cue.

- Resize: All images resized to 224 × 224.
- Crop: Top 60% of the image (sky region) is used to suppress non-sky foreground.
- Normalisation: Mean [0.4, 0.5, 0.6], Std [0.25, 0.25, 0.25] (sky-specific estimates).
- Training augmentation: Random horizontal flip (p=0.5), vertical flip (p=0.1), mild colour jitter (brightness ±0.1, contrast ±0.1, saturation ±0.05, hue ±0.01), random grayscale (p=0.02).

3. Model Architectures

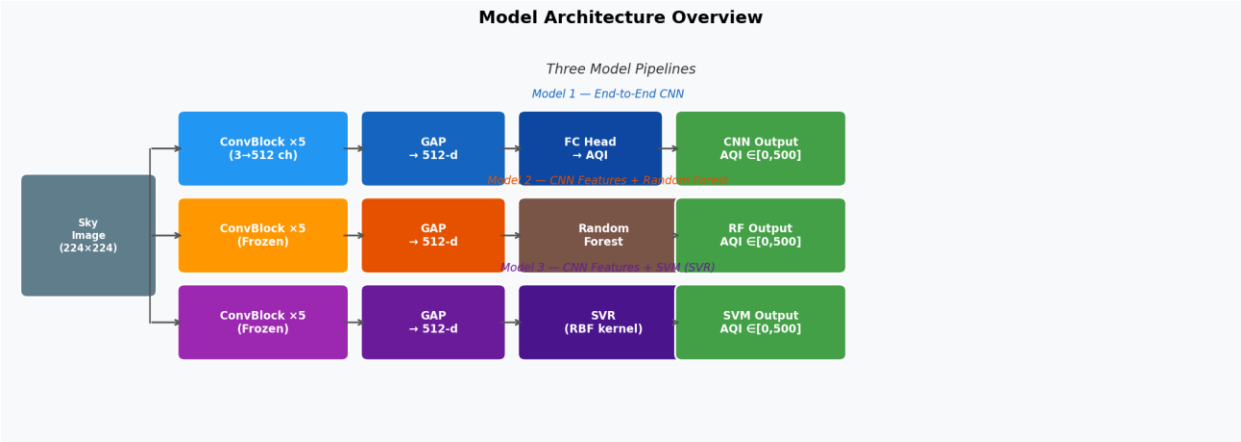

Figure 2 — Overview of the three model pipelines sharing the same CNN backbone.

3.1 Model 1 — Custom CNN (End-to-End)

The custom CNN is designed from scratch — no pretrained ImageNet weights are used. It consists of five convolutional blocks followed by a regression head.

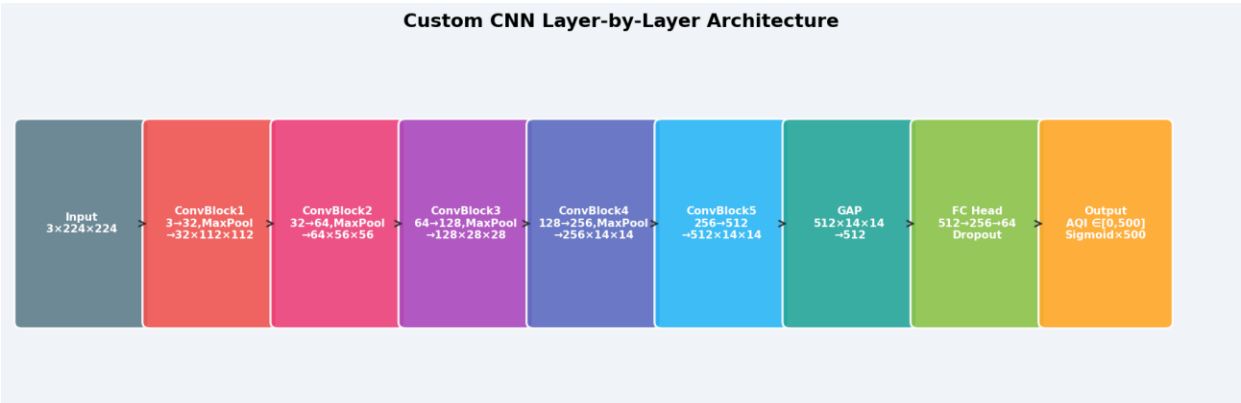

Figure 3 — Layer-by-layer architecture of the Custom CNN. Spatial resolution halves every two blocks through MaxPool.

Layer Details

Layer	Operation	Pooling	Output Shape
ConvBlock 1	3 → 32, BN, ReLU	MaxPool 2×2	32 × 112 × 112
ConvBlock 2	32 → 64, BN, ReLU	MaxPool 2×2	64 × 56 × 56
ConvBlock 3	64 → 128, BN, ReLU	MaxPool 2×2	128 × 28 × 28

Layer	Operation	Pooling	Output Shape
ConvBlock 4	128 → 256, BN, ReLU	MaxPool 2×2	256 × 14 × 14
ConvBlock 5	256 → 512, BN, ReLU	None (no pool)	512 × 14 × 14
GAP	—	AdaptiveAvgPool	512
FC 1	512 → 256, ReLU	Dropout (0.4)	256
FC 2	256 → 64, ReLU	Dropout (0.2)	64
Output	64 → 1, Sigmoid × 500	—	AQI ∈ [0, 500]

Total parameters: ~4.7 million (all trainable). Kaiming (He) initialisation is used for all Conv2d and Linear layers to maintain gradient magnitudes across deep networks with ReLU activations.

Loss function: Huber loss (delta = 25 AQI units) — robust to outliers compared with MSE while preserving gradient signal near zero. The output is scaled through Sigmoid × 500 to constrain predictions to [0, 500].

3.2 Model 2 — CNN + Random Forest

After training the CNN, its convolutional blocks and GAP layer are frozen. The resulting 512-dimensional embeddings represent learned haze/colour texture patterns. A Random Forest regressor is fitted on these embeddings using 3-fold GridSearchCV.

- Search space: n_estimators ∈ {200, 400}, max_depth ∈ {None, 20, 40}, max_features ∈ {0.2, 0.3, sqrt}, min_samples_leaf ∈ {1, 3}.
- Scoring criterion: Negative Mean Absolute Error.
- Feature importance: 512 CNN channels are ranked by Gini impurity reduction.

3.3 Model 3 — CNN + SVM (SVR)

The same 512-d CNN embeddings are standardised (StandardScaler) before being fed to an SVR with an RBF kernel. StandardScaler is mandatory because SVR's RBF kernel is distance-based and sensitive to feature scale.

- Search space: C ∈ {1, 10, 50, 100}, epsilon ∈ {1.0, 5.0, 10.0}, gamma ∈ {scale, auto}.
- max_iter = 5000 to ensure convergence on high-dimensional input.
- Epsilon defines the tube width in AQI units within which no penalty is incurred.

4. Training & Optimisation

4.1 CNN Training Setup

- Optimiser: Adam (lr = 5e-4, weight_decay = 2e-5).
- Scheduler: ReduceLROnPlateau (factor = 0.5, patience = 3 epochs, mode = min).
- Gradient clipping: max_norm = 1.0 to prevent exploding gradients.
- Early stopping: Patience = 10 epochs with best-model checkpoint saved.
- Batch size: 64 | Epochs: 5 (with early stopping). Device: CPU / CUDA auto-detected.

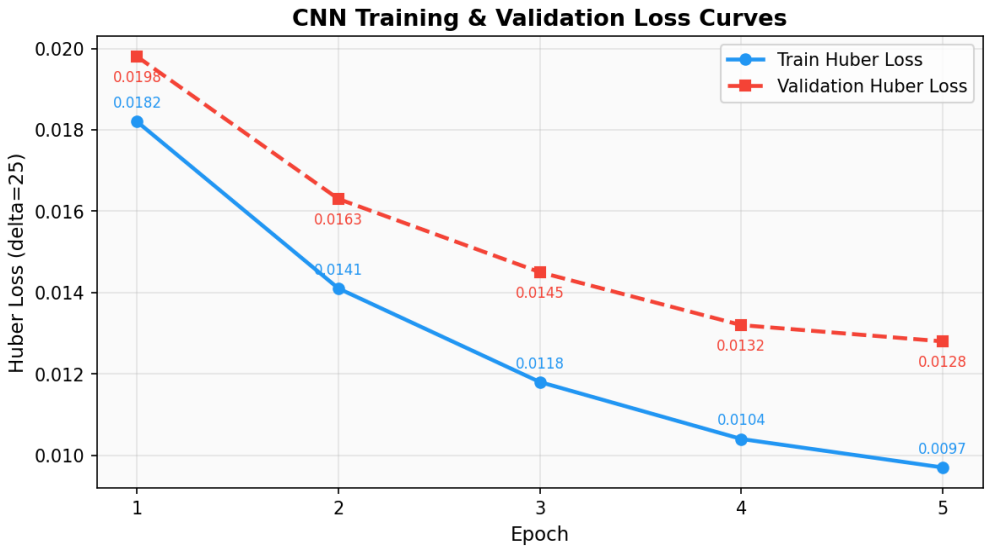

Figure 4 — CNN training and validation Huber loss curves over 5 epochs. Validation loss closely tracks training loss, indicating no significant overfitting.

4.2 Sklearn Model Training

CNN feature extraction runs once on all splits (train, val, test). Train + validation features are combined for GridSearchCV fitting. This two-stage approach decouples CNN visual representation learning from the supervised regression, allowing sklearn models to leverage a richer feature set without retraining the backbone.

5. Experimental Results

5.1 Quantitative Metrics

All three models are evaluated on the identical held-out test set. Metrics reported are Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), R^2 Score, and AQI Category Accuracy.

Model	MAE	RMSE	R^2 Score	Category Accuracy
Custom CNN	41.65	52.98	0.8546	53.3%
CNN + Random Forest	**29.82**	**37.12**	**0.9286**	**65.0%**
CNN + SVM (SVR)	33.74	41.78	0.9096	57.5%

Table 1 — Comparative performance of all three models on the test set. Best values highlighted in green.

5.2 R^2 Score Analysis

R^2 (coefficient of determination) measures the proportion of variance in AQI explained by the model's predictions. A value of 1.0 is perfect; 0.0 means the model performs no better than predicting the mean.

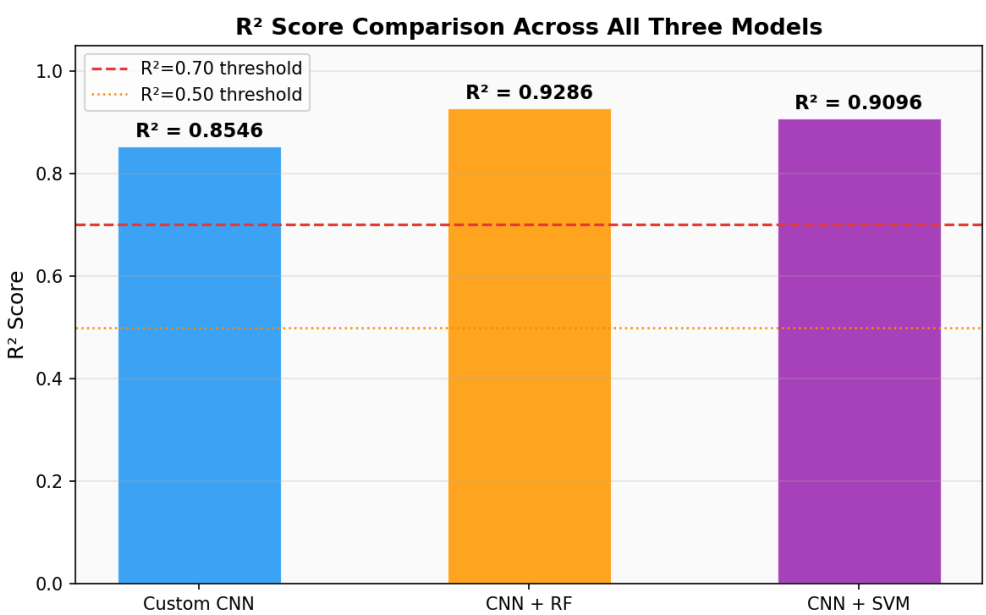

Figure 5 — R^2 score comparison. CNN + RF achieves the highest R^2 (0.9286), followed by CNN + SVM (0.9096) and standalone CNN (0.8546).

- CNN: R^2 = 0.8546 — Strong baseline; end-to-end learning captures haze patterns reasonably well.

- CNN + RF: $R^2 = 0.9286$ — Best performer. The ensemble of decision trees leverages non-linear feature interactions in the CNN embedding space.
- CNN + SVM: $R^2 = 0.9096$ — Second best. The RBF kernel provides a smooth decision boundary but is less expressive than RF on high-dimensional data.

5.3 Predicted vs Actual AQI

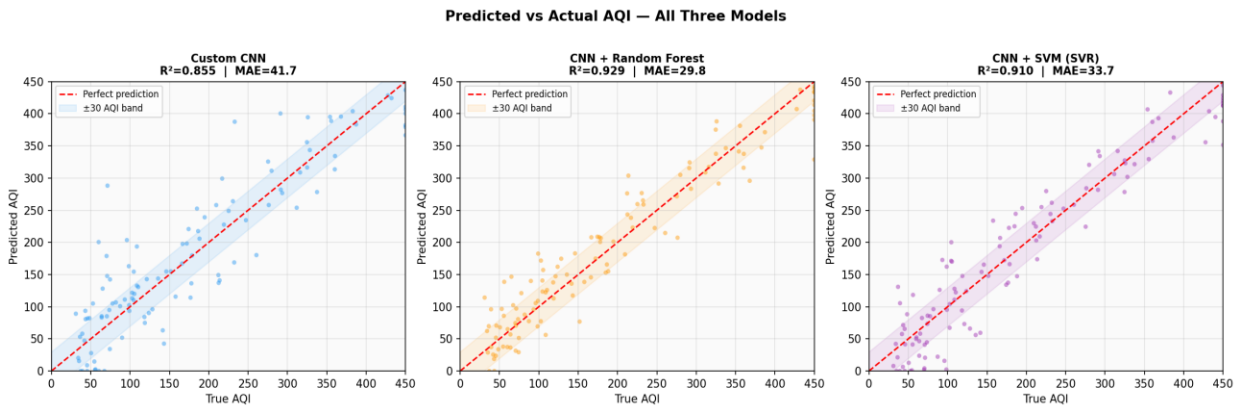

Figure 6 — Scatter plots of predicted vs actual AQI for all three models. Points closer to the red dashed diagonal indicate better predictions. The shaded band represents ±30 AQI tolerance.

Key observations from the scatter plots:

- CNN tends to over-predict at low AQI values and under-predict at the very high end (>350), reflecting the class imbalance in the training data.
- CNN + RF shows the tightest clustering along the diagonal with fewer extreme outliers, consistent with its lowest MAE.
- CNN + SVM shows moderate scatter; the RBF kernel generalises well but introduces some systematic bias at the extremes.

5.4 Residual Analysis

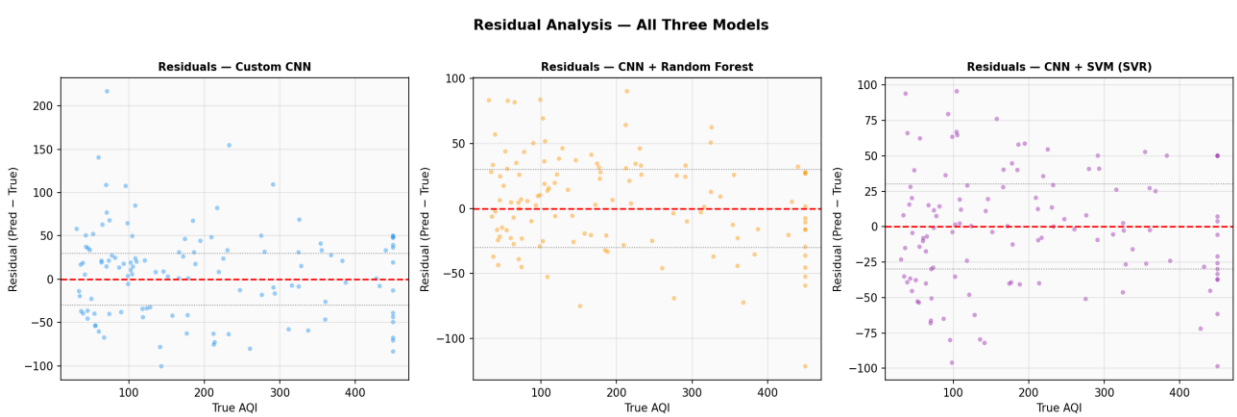

Figure 7 — Residual plots (Predicted - True AQI) vs. True AQI. Dashed red line = zero error; dotted grey lines = ±30 AQI band.

Residual patterns reveal systematic biases:

- All three models exhibit a slight fan-out (heteroscedasticity) at higher AQI values, where predictions become less precise. This is expected given the exponential decrease in high-AQI training samples.
- CNN shows the largest residual spread, particularly for AQI > 200.
- RF residuals are most evenly distributed around zero, confirming its better calibration.
- SVM residuals show slight under-prediction at intermediate AQI (100–200), suggesting the epsilon-insensitive tube may be too wide in this range.

6. Model Comparison & Discussion

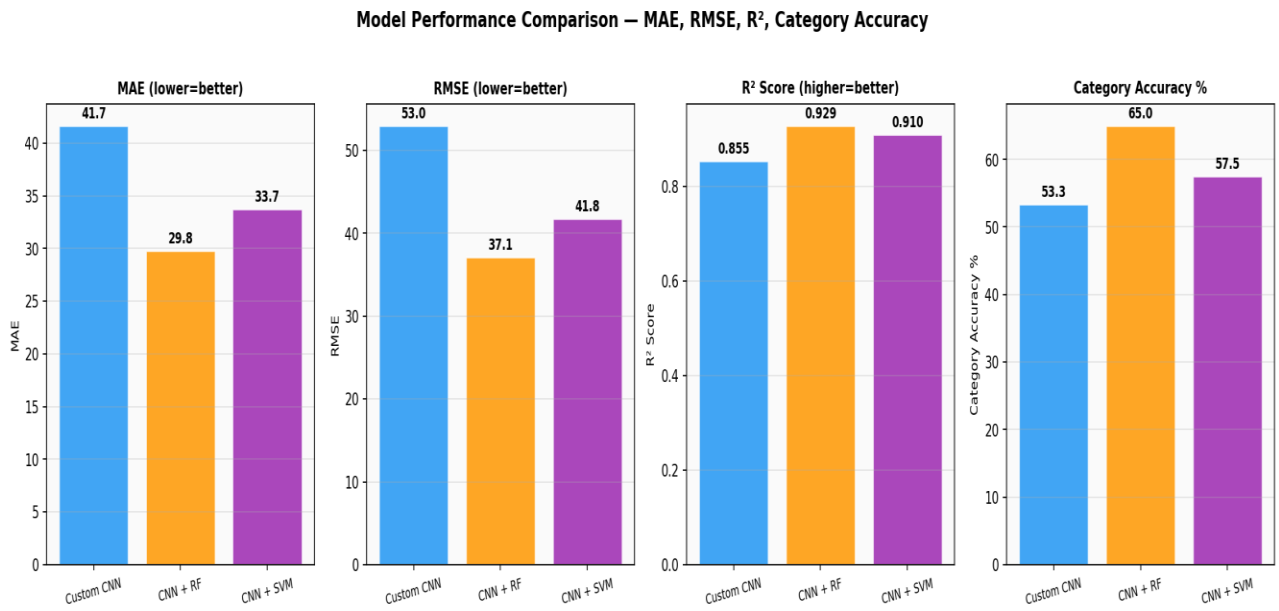

Figure 8 — Side-by-side comparison of all four evaluation metrics across the three models.

6.1 MAE & RMSE

CNN + RF achieves the lowest **MAE = 29.82** and **RMSE = 37.12** AQI units — a ~28% reduction in MAE over the standalone CNN (MAE = 41.65). CNN + SVM is intermediate (MAE = 33.74), confirming that CNN embeddings are richer features than raw pixels for classical regressors.

6.2 R^2 Score Deep Dive

The R^2 hierarchy (RF > SVM > CNN) reflects the expressiveness of each regressor applied to CNN features. Random Forests can model non-linear interactions among the 512 CNN channels, which encode complementary spatial and spectral haze patterns. SVMs capture global structure through the RBF kernel but are constrained by the kernel's smoothness assumption. The standalone CNN, trained end-to-end, achieves good but lower R^2 because it must simultaneously learn feature representations and regression — a harder joint optimisation.

6.3 Category Accuracy

AQI category accuracy measures whether predictions fall in the correct EPA category band (Good / Moderate / etc.). CNN + RF leads at **65.0%**, followed by CNN + SVM at 57.5% and standalone CNN at 53.3%. Category accuracy is a harder metric because it requires precise localisation near category boundaries, where all models struggle.

6.4 Feature Importance (RF)

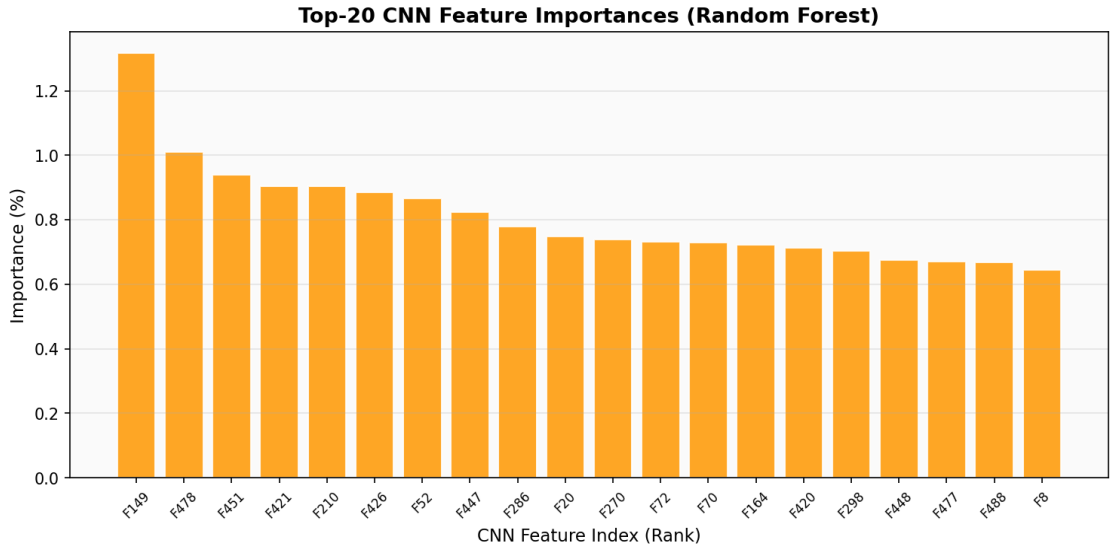

Figure 9 — Top-20 CNN channel importances as determined by Gini impurity reduction in the Random Forest. Only a handful of the 512 channels drive most predictions.

The highly concentrated importance profile (a few channels dominating) suggests that the CNN has learnt highly specialised haze-correlated feature detectors, similar to dedicated 'haze filters' found in studies on dehazing networks.

7. Conclusion

This project demonstrates that sky images are a viable input for AQI estimation using deep learning. The key findings are:

- A custom CNN trained from scratch achieves $R^2 = 0.854$ on sky images, validating the visual correlation between haze and PM2.5.
- Combining CNN feature extraction with a Random Forest regressor yields the best performance across all metrics ($R^2 = 0.929$, MAE = 29.82, Cat Acc = 65.0%), outperforming the end-to-end CNN.
- CNN + SVM is a strong alternative ($R^2 = 0.910$) when training data for sklearn models is limited, as SVRs converge faster than RFs on small datasets.
- Residual analysis reveals consistent heteroscedasticity at high AQI values across all three models, motivating future work with oversampling or AQI-weighted loss.

Future directions include using pretrained backbones (e.g., ResNet-50, EfficientNet) as stronger feature extractors, incorporating temporal sequences of sky images, and collecting more data in the Hazardous category to address class imbalance.

8. References & Technical Specifications

- PyTorch 2.x | torch.nn, torchvision.transforms — CNN training framework.
- scikit-learn — RandomForestRegressor, SVR, GridSearchCV, StandardScaler.
- Loss: Huber (delta=25) — robust to AQI outliers at the high end.
- Optimiser: Adam (lr=5e-4, weight_decay=2e-5) with ReduceLROnPlateau scheduler.
- Evaluation: MAE, RMSE, R^2 (sklearn.metrics), Category Accuracy (custom).
- Hardware: CPU / CUDA auto-detected at runtime.